

Lecture on Rheumatic Fever

Learning Objectives

- Define rheumatic fever and its etiology.
- Describe the pathophysiology and Jones criteria for diagnosis.
- Identify clinical manifestations and complications.
- Discuss management, prevention, and nursing care for patients with rheumatic fever.

Introduction

Rheumatic fever (RF) is an inflammatory disease that may develop after an infection with Group A beta-hemolytic streptococcus (GAS). It primarily affects the heart, joints, skin, and central nervous system. The condition is most common among children aged 5–15 years.

Etiology and Pathophysiology

Rheumatic fever is a delayed autoimmune response to pharyngeal infection with Group A streptococci. Antibodies produced against streptococcal antigens cross-react with human tissues, leading to inflammation in the heart (pancarditis), joints, brain, and skin.

[Insert Diagram: Pathophysiology of Rheumatic Heart Disease]

Clinical Features

The clinical presentation of rheumatic fever varies, but major manifestations are summarized in the Jones criteria.

Major Criteria:

- Carditis
- Polyarthrititis
- Chorea (Sydenham's chorea)
- Erythema marginatum
- Subcutaneous nodules

Minor Criteria:

- Fever
- Arthralgia
- Elevated ESR or CRP
- Prolonged PR interval on ECG

[Insert Table or Diagram: Jones Criteria for Diagnosis]

Diagnosis

Diagnosis of rheumatic fever is based on the revised Jones criteria, which require evidence of a preceding streptococcal infection along with the presence of major and minor criteria.

Management

Management involves eradication of streptococcal infection, control of inflammation, and prevention of recurrence.

1. Antibiotic therapy: Benzathine penicillin G (single intramuscular injection) is the drug of choice.
2. Anti-inflammatory therapy: Aspirin or corticosteroids to control inflammation.
3. Supportive care: Bed rest, adequate nutrition, and fluid balance.
4. Long-term prophylaxis: Monthly benzathine penicillin injections for at least 5–10 years or until adulthood.

Nursing Care

Nursing management focuses on relieving symptoms, preventing complications, and promoting recovery. Key responsibilities include:

- Monitoring vital signs and cardiac function
- Administering prescribed medications and antibiotics
- Educating the patient and family on adherence to prophylaxis
- Encouraging rest during acute phases
- Promoting good oral and throat hygiene to prevent infections

Prevention

Primary prevention involves prompt treatment of streptococcal throat infections.

Secondary prevention focuses on long-term antibiotic prophylaxis to prevent recurrence.

Summary

Rheumatic fever is a serious post-streptococcal autoimmune condition that can lead to chronic rheumatic heart disease if untreated. Early diagnosis, appropriate management, and preventive strategies are essential to reduce morbidity among children and adolescents.